

NATIONAL UNIVERSITY OF SINGAPORE

EXAMINATION FOR
(Semester II: 2021/2022)

EE5507 – ANALOG INTEGRATED CIRCUIT DESIGN

Apr / May 2022 - Time Allowed: 2.5 Hours

INSTRUCTIONS TO CANDIDATES:

1. This paper contains **FOUR** (4) questions and comprises **EIGHT** (8) printed pages.
2. Answer all **FOUR** (4) questions.
3. Each question carries 25 marks.
4. This is an OPEN BOOK examination.
5. Calculators can be used in the examination.
6. You may assume the following constant:
 $q=1.6021765 \times 10^{-19} \text{ C}$ $k=1.38064852 \times 10^{-23} \text{ m}^2 \text{ kg s}^{-2} \text{ K}^{-1}$

Q.1

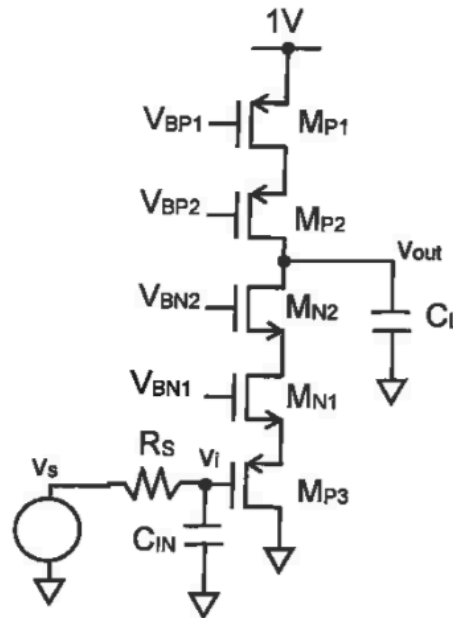


Fig. Q1

In Fig. Q1, v_s is a sensor with high R_S of $100\text{ k}\Omega$ and DC voltage of 0 V . A single-ended to differential low noise amplifier (LNA) is shown in Fig. Q1. Assume all the NMOS and PMOS body are connected to their source terminal respectively (No body effect). The supply voltage is 1 V and the whole circuit consume $50\text{ }\mu\text{A}$. The device parameters are given as follows:

- $L=130\text{ nm}$: $V_{\text{THN}}=0.24\text{ V}$, $\mu_n C_{\text{ox}}=380\text{ }\mu\text{A/V}^2$, $\lambda_n=0.8\text{ V}^{-1}$.
- $L=130\text{ nm}$: $V_{\text{THP}}=-0.24\text{ V}$, $\mu_p C_{\text{ox}}=110\text{ }\mu\text{A/V}^2$, $\lambda_p=0.8\text{ V}^{-1}$.
- $(W/L)_{\text{MN1}}=(W/L)_{\text{MN2}}=1.3\text{ }\mu\text{m}/0.13\text{ }\mu\text{m}$, and $(W/L)_{\text{MP1}}=(W/L)_{\text{MP2}}=2.6\text{ }\mu\text{m}/0.13\text{ }\mu\text{m}$
- $(W/L)_{\text{MP3}}=5.2\text{ }\mu\text{m}/0.13\text{ }\mu\text{m}$

- Determine $g_{m,\text{MN1}}$, $g_{m,\text{MN2}}$, $g_{m,\text{MP1}}$, $g_{m,\text{MP2}}$, $r_{o,\text{MN1}}$, $r_{o,\text{MN2}}$, and $r_{o,\text{MP1}}$, $r_{o,\text{MP2}}$. (6 marks)
- Determine the driving point resistance looking into the drain terminal of MP_2 and drain terminal of MN_2 . (6 marks)
- Determine the driving point resistance looking into the source terminal of MN_1 . (4 marks)
- Determine v_{out}/v_s . (**Hint:** the amplifier is $\text{CD} \rightarrow \text{CG} \rightarrow \text{CG}$). (5 marks)
- The dominant poles are mainly determined by C_L and C_S . Explain the reason. (4 marks)

Q.2

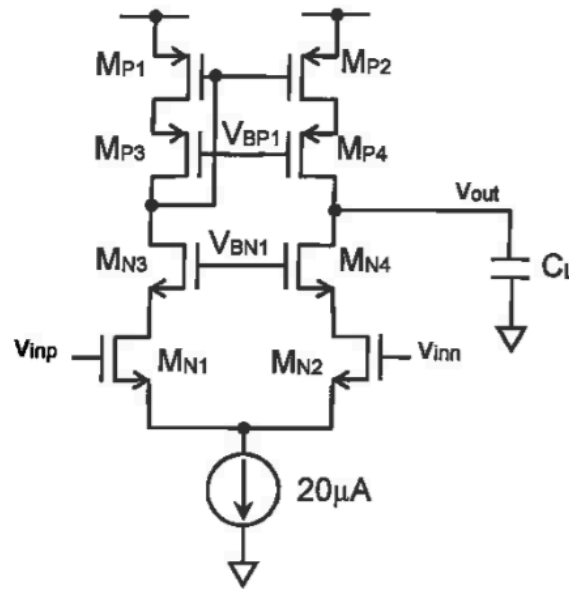


Fig. Q2a

Fig. Q2a shows a telescopic opamp with supply voltage of 1V. You may assume the following:

- NMOS: $V_{THN}=0.24V$, $\mu_n C_{ox}=380\mu A/V^2$, $\lambda_n=0.8V^{-1}$.
- PMOS: $V_{THP}=-0.24V$, $\mu_p C_{ox}=110\mu A/V^2$, $\lambda_p=0.8V^{-1}$.
- $(W/L)_{MN1}=(W/L)_{MN2}=(W/L)_{MN3}=(W/L)_{MN4}=1.3\mu m/130nm$.
- $(W/L)_{MP1}=(W/L)_{MP2}=(W/L)_{MP3}=(W/L)_{MP4}=2.6\mu m/130nm$.
- $M_{N1}, M_{N2}, M_{N3}, M_{N4}$: $g_{mn}=276\mu A/V$ and $r_{on}=125k\Omega$.
- $M_{P1}, M_{P2}, M_{P3}, M_{P4}$: $g_{mp}=210\mu A/V$ and $r_{op}=125k\Omega$.

a) Estimate the overall gain, i.e. $(V_{out})/(V_{inp}-V_{inn})$.

(5 marks)

b) Determine C_L such that unity gain bandwidth is 0.5MHz.

(4 marks)

c) Determine the slew rate of the opamp.

(5 marks)

d) Determine V_{BP1} that will maximize v_{out} range.

(4 marks)

- e) The telescopic opamp is used for the inverting amplifier shown in Fig. Q2b. Determine V_{BN1} that will maximize v_{out} range.

(4 marks)

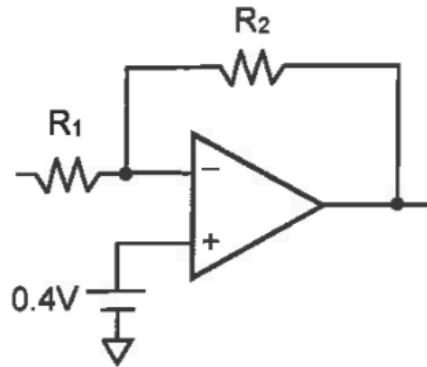


Fig. Q2b

- f) Based on (d) and (e), show that the maximum signal frequency that the inverting amplifier in Fig. Q2b can operate is not limited by slew rate.

(3 marks)

Q3.

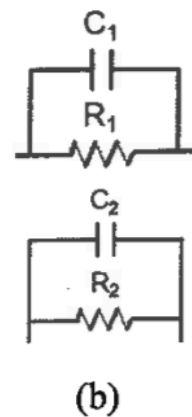
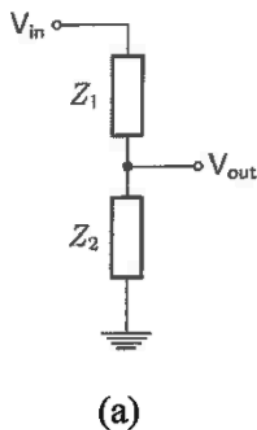


Fig. Q3

a) Consider the bilinear circuit shown in Fig. Q3(a) where Z_1 and Z_2 are parallel connection of resistor and capacitor as shown in Fig. Q3(b).

i) Derive the transfer function $T(s) = V_{out}/V_{in}$.

(5 marks)

ii) For the 3 cases below, find DC gain $T(0)$ and high frequency gain $T(\infty)$.

(5 marks)

	Case 1	Case 2	Case 3	
R_1	100	100	100	Ω
R_2	200	200	200	Ω
C_1	2	1	4	nF
C_2	1	1	1	nF
$T(0)$				V/V
$T(\infty)$				V/V

iii) Sketch Bode plot of $|T(j\omega)|$ and indicate ω_z and ω_p for the 3 cases.

(5 marks)

	Case 1	Case 2	Case 3	
ω_z				Mrad/s
ω_p				Mrad/s

iv) For case 2 sketch the time domain response $V_{out}(t)$ for $V_{in}(t)$ shown in Fig. Q3(c).
(Hint: Consider $T(0)$ and $T(\infty)$. Any time constants should be defined.)

(5 marks)

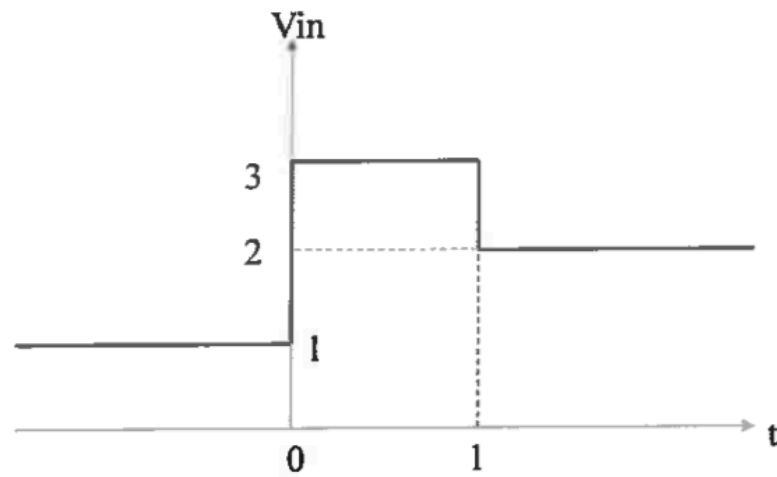


Fig. Q3 (c)

b) For each of the statements below regarding delta-sigma ADC, provide the answer of **TRUE** (if the statement is correct) or **FALSE** (if the statement is wrong) in the answer booklet.

(Marks will be deducted for each wrong answer you give. No mark will be deducted if you do not answer.)

i) Delta-sigma converters do not use oversampling.

TRUE / FALSE
(1 mark)

ii) Delta-sigma converters use noise shaping.

TRUE / FALSE
(1 mark)

iii) Delta-sigma modulators are intrinsically nonlinear because of quantizer.

TRUE / FALSE
(1 mark)

iv) As the delta-sigma modulator order increases, the in-band quantization noise power increases.

TRUE / FALSE
(1 mark)

v) Single bit quantizers are inherently linear because of 2 output levels.

TRUE / FALSE
(1 mark)

Q4.

a) Consider the auto-zeroed amplifier shown in Fig. Q4:

i) If the amplifier's offset $V_{os} = 10\text{mV}$, determine the gain A of the basic amplifier such that the input-referred offset after auto-zeroing is less than $1\mu\text{V}$.

(5 marks)

ii) If both the auto-zeroing and the amplification phases last $10\mu\text{s}$, and given that all the switches have an on-resistance of $2\text{k}\Omega$, determine the worst-case value of C_1 (same as C_2) such that settling errors are below 0.1% (input-referred). [Note: the opamp may be assumed to be ideal.]

(5 marks)

iii) If both the auto-zeroing and the amplification phases last $10\mu\text{s}$, determine the unity-gain bandwidth of the opamp such that all settling errors are below 0.1% (input-referred). [Note: All switches may be assumed to be ideal.]

(5 marks)

iv) If the amplifier is assumed to be noise-free, what value of C_1 (same as C_2) ensures that the residual input-referred noise after auto-zeroing is less than $1\mu\text{V}$ (rms).

[Note: $4kT = 1.62 \times 10^{-20}$ (W/Hz)]

(5 marks)

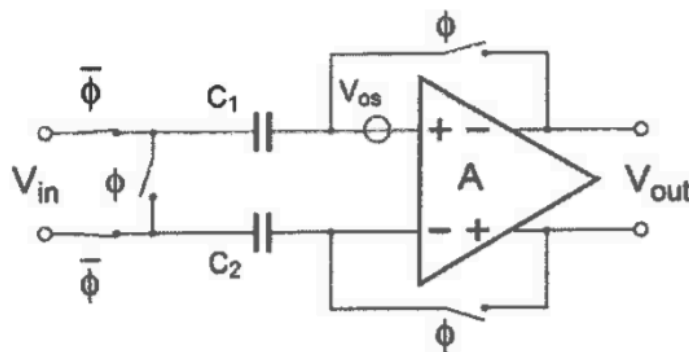


Fig. Q4

b) For each of the statements below regarding a current steering DAC, provide the answer of **TRUE** (if the statement is correct) or **FALSE** (if the statement is wrong) in the answer booklet.

(Marks will be deducted for each wrong answer you give. No mark will be deducted if you do not answer.)

i) For dynamic performance: Moderate swing steering switch driver is worse than CMOS rail to rail driver.

TRUE / FALSE
(1 mark)

ii) For analog current consumption: Moderate swing steering switch driver is worse (larger) than CMOS rail to rail driver.

TRUE / FALSE
(1 mark)

iii) For dynamic performance: dynamic errors are due to transient DAC output current changes caused by transient voltage changes at the current source output nodes due to coupling from driver device gate voltage change.

TRUE / FALSE
(1 mark)

iv) For dynamic performance: the pair of PMOS (or NMOS) current steering switches can be OFF at the same time.

TRUE / FALSE
(1 mark)

v) Dynamic performance can be improved using $R_F C_F$ filtering in the DAC amplifier stage

TRUE / FALSE
(1 mark)